panel



File: panel_main.kicad_sch

awong-dev/pager

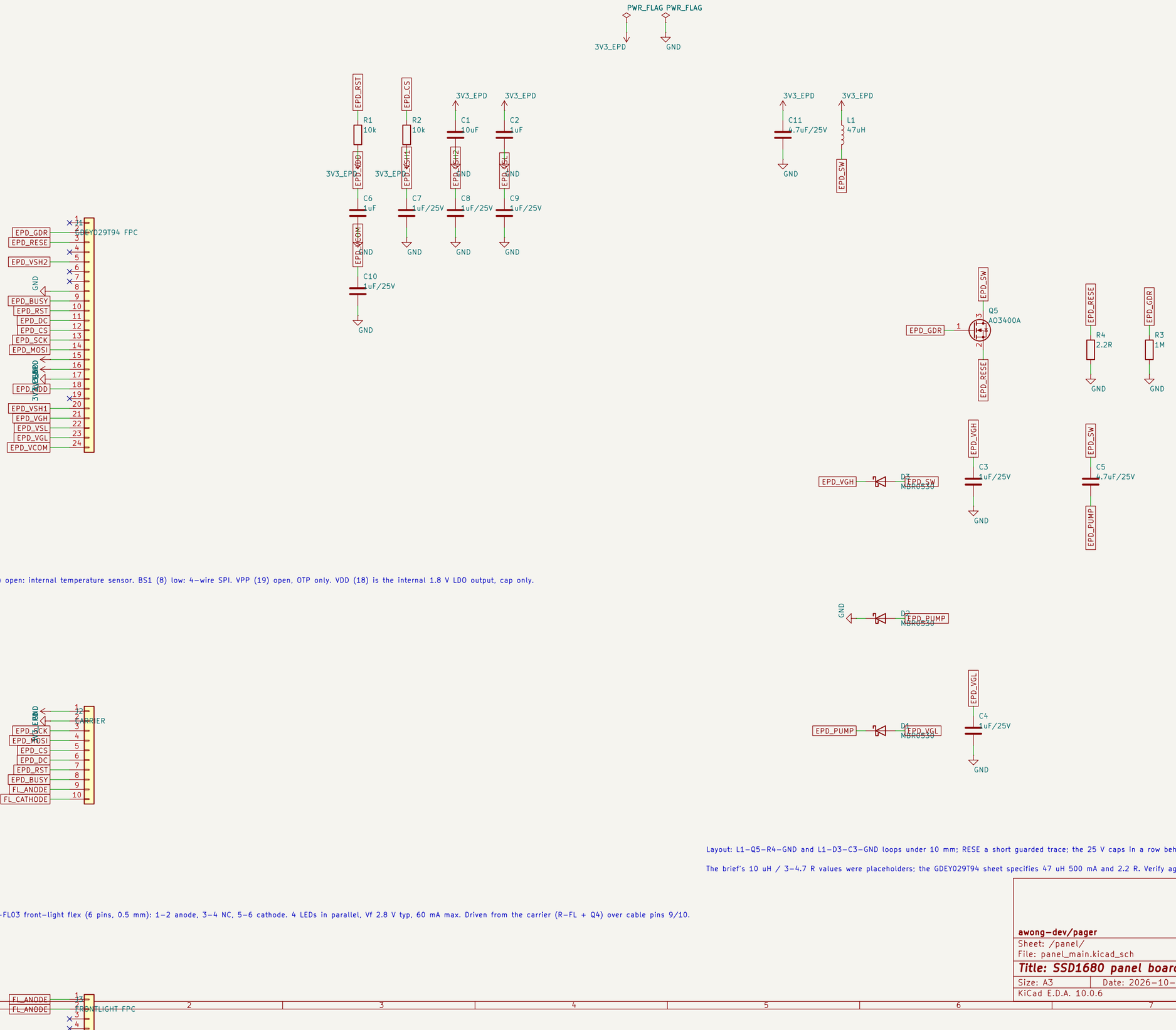
Sheet: /

File: pager-panel.kicad_sch

Title: pager-panel

Size: A4	Date: 2026-10-17	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/2

Panel board for the Good Display GDEY029T94 (2.9" 296x128, SSD1680). Reference circuit from the GDEY029T94 specification (24.10.22): RESE 2.2R, L 47uH, MBR0530 x3, 25 V caps. 3V3_EPD -> L1 -> SW node; Q5 (gate GDR) switches SW to GND through R4 (RESE sense). D3 SW->VGH into C3. C5 flies SW->PUMP; D2 clamps PUMP to GND (a



TSCL/TSDA (6/7) open: internal temperature sensor. BS1 (8) low: 4-wire SPI. VPP (19) open, OTP only. VDD (18) is the internal 1.8 V LDO output, cap only.

J3: GDEY029T94-FL03 front-light flex (6 pins, 0.5 mm): 1-2 anode, 3-4 NC, 5-6 cathode. 4 LEDs in parallel, Vf 2.8 V typ, 60 mA max. Driven from the carrier (R-FL + Q4) over cable pins 9/10.

Layout: L1-Q5-R4-GND and L1-D3-C3-GND loops under 10 mm; RESE a short guarded trace; the 25 V caps in a row behind J1 pins 18-24; ground pour under the cluster.

The brief's 10 uH / 3-4.7 R values were placeholders; the GDEY029T94 sheet specifies 47 uH 500 mA and 2.2 R. Verify against the sheet for the exact panel ordered.

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Sheet: /panel/

File: panel_main.kicad_sch

Title: SSD1680 panel board: ZIF, boost and charge pump, cable to carrier

Size: A3	Date: 2026-10-17	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 2/2